Data-quality-aware human deferral for cybersecurity alert triage under analyst capacity constraints

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Short communication.

Abstract

Knowledge-based triage must allocate scarce analyst capacity when telemetry quality deteriorates. We investigate whether observable quality improves human deferral beyond detector probabilities, separating estimation from mandatory escalation and review volume. A frozen analysis plan evaluates label-independent field-family outages on verified, deduplicated UNSW-NB15 and temporally separated CICIDS2017 corpora, using two explicitly simulated reviewer scenarios. In the primary UNSW comparison at 20% capacity, quality information reduced expected cost by 3.20% relative to probability-only estimation, with identical review counts and mandatory ordering. The absolute reduction was 0.00686; the 95% Monte Carlo interval was 0.00561-0.00811 over 20 experimental randomizations. These intervals condition on the fixed benchmark corpora. The CIC outage comparison did not corroborate the benefit, and confidence routing achieved lower mean cost there. Under severe outages, mandatory review covered every flow, leaving 80% unresolved and making the automation-head contrast structurally zero. A capacity-constrained conformal comparator exposed a different balance between automatic errors and unresolved cases. The study identifies conditional estimation benefits and capacity-driven failure boundaries for AI governance; it does not establish analyst performance or production security effectiveness.

Keywords: Knowledge-based systems; Cybersecurity; Learning to defer; Data quality; AI governance; Analyst capacity; Managerial decision-making

1 Introduction

Security operations centers combine people, processes and technology [1]. Data governance assigns responsibilities and controls [2,3], while mandatory human oversight alone does not demonstrate effectiveness [4]. For a manager, the relevant question is which decisions require scarce review and what happens when that review is unavailable.

Intrusion-detection research offers extensive machine-learning and deep-learning methods [5,6,7]. Dataset fidelity and transfer remain separate challenges [8,9,10,11]. A confident detector can therefore provide an inadequate basis for triage when its inputs or deployment environment change.

Rejection and learning to defer formalize selective prediction and expert referral [12,13,14]; DeCCaF already combines asymmetric costs and workload constraints [15]. SOC-specific learning to defer with human feedback

also models analyst availability [16]. Dual-threshold conformal triage separates automatic closure, escalation and deferral [17]. We claim neither the first cybersecurity deferral system nor a new consistent surrogate.

Data preparation, completeness and distributional quality can affect learned predictions [18,19,20]. Uncertainty and concept-drift research motivate testing their relationship to observable quality rather than equating them [21,22]. Human-in-the-loop and knowledge-based decision support connect predictions to organizational action [23,24,25]. We ask: **when does telemetry quality improve triage beyond predictive uncertainty under a fixed analyst budget?** Our contribution is a reproducible conditions analysis, with matched loss-estimator comparisons, a temporal replication and explicit accounting for unresolved attacks.

2 Method

2.1 Knowledge rules and allocation

Each completed flow is a proxy alert unit. Let p_i be calibrated attack probability, u_i missing-field fraction and v_i training-range outlier fraction. Bounds are $[Q_{.01,j} - 0.1t_j, Q_{.99,j} + 0.1t_j]$, $t_j = \max(Q_{.99,j} - Q_{.01,j}, 1)$, fitted on training rows. These proxies can flag valid extreme traffic. The experimental policy sets

$$\hat{y}_i = \mathbf{1}[p_i \geq 1/6], \qquad m_i = \mathbf{1}[u_i \geq 0.5 \ \vee \ v_i \geq 0.25]. \tag{1}$$

The knowledge base prohibits automation when $m_i = 1$. These thresholds are experimental policy assumptions. Costs are missed attack 5, benign escalation 1, review $c_R = 0.05$ and unresolved hold $c_H = 1.5$. For binary automate, review and hold indicators a_i, r_i, h_i, estimated cost is

$$\widehat{J} = \frac{1}{n} \sum_i \left[a_i \widehat{\ell}_i^A + r_i (\widehat{\ell}_i^H + c_R) + h_i c_H \right]. \tag{2}$$

Require $a_i + r_i + h_i = 1$, $\sum_i r_i \leq B = \lfloor bn \rfloor$, $a_i \leq 1 - m_i$ and $h_i \leq m_i$. With $M = \{i : m_i = 1\}$, exactly $\max(|M| - B, 0)$ cases are held. This imposes mandatory service priority. It is not an unconstrained economic optimum.

$$\widehat{g}_i = \widehat{\ell}_i^A - \widehat{\ell}_i^H - c_R. \tag{3}$$

Allocation property. If $|M| \leq B$, review all mandatory cases, then optional cases with the largest positive gains in Eq. (3). Under overload, review the B mandatory cases with smallest estimated reviewer losses. An exchange argument proves fixed-estimated-loss optimality: optional review changes cost by $-\widehat{g}_i/n$; the constant hold term cancels under overload. Sorting costs $O(n \log n)$. Changing only the automation head cannot affect overloaded routing when the reviewer head is fixed.

The matched arm fills optional review slots, including nonpositive estimated gains, and shares mandatory ordering. It controls review volume; native arms use positive-gain admission. Figure 1 separates estimation, policy and capacity.

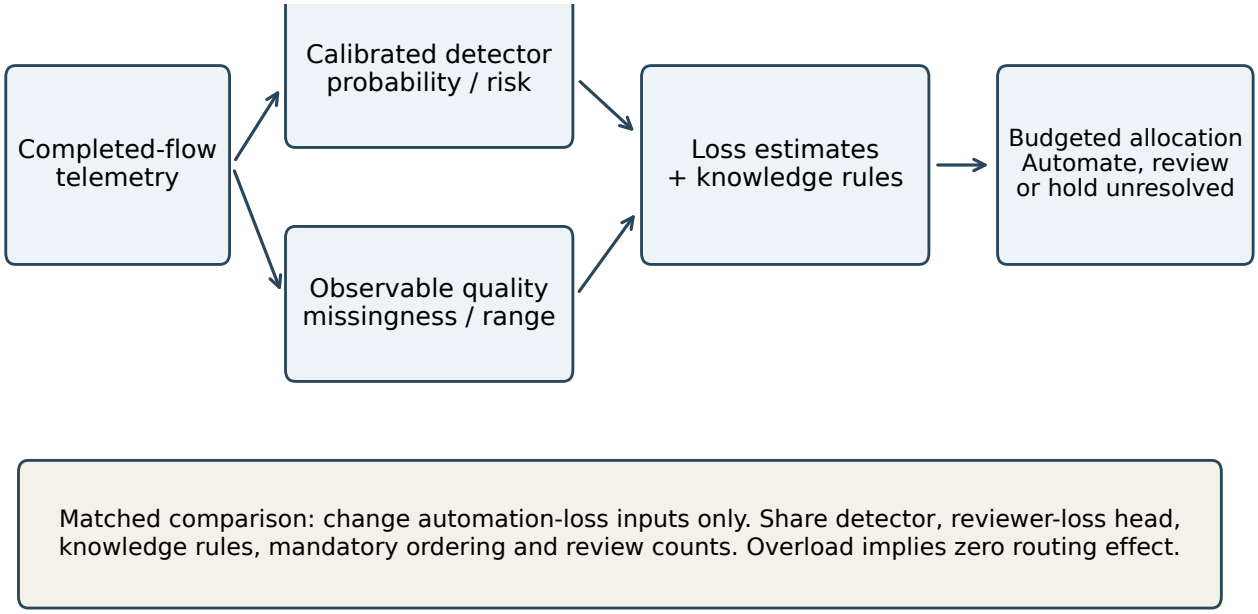


Figure 1: Knowledge-based triage and matched comparison. Only automation-loss inputs change in the primary contrast; detector, reviewer estimates, rules, ordering and review counts are common. Holds remain unresolved. Programmatic diagram with AI-assisted coding.

2.2 Loss estimation and reviewer assumptions

Training targets are $L_i^A = w(y_i)\mathbf{1}[\hat{y}_i \neq y_i]$ and $L_i^H = w(y_i)\mathbf{1}[\text{review}_i \neq y_i]$, with $w(y) = 1+4y$. Probability-only heads receive p_i and $\min(5p_i, 1-p_i)$. Quality heads add u_i, v_i, four family fractions and their interactions with p_i. The observed-feature comparator contains that entire representation plus raw observed features. Every head receives the same two-option search. All gain rows have outcome labels; reviewer judgments are logged randomly with probability 0.4. No selective-feedback identification claim is made.

No analyst judgments were available. One simulated reviewer has constant error 0.10; another has error

$$e_i = \min\{0.45,\ 0.05 + 0.25\mathbf{1}[\hat{y}_i \neq y_i] + 0.1(1 - |2p_i^{clean} - 1|)\}. \tag{4}$$

Methods share simulated fitting judgments and expected evaluation losses. Reviewer error is not defined from quality features. Its clean benchmark counterpart is a latent simulation input unavailable to the router. Human skill, fatigue and investigation time are unmeasured.

3 Experimental design

3.1 Verified corpora and split isolation

UNSW-NB15 provides cyber-range traffic and official development/test partitions [26,27]; its associated anomaly-detection and feature-generation literature supplies dataset context [28,29,30]. CICIDS2017 provides scheduled benign and attack traffic [31]. Known benchmark construction problems motivate explicit verification [32]. We use pinned mirrors with matching mirror-published SHA256 values because direct provider downloads were inaccessible or required a personal-information form. These checks verify acquired bytes and metadata, not original ground truth.

UNSW source partitions contain 175,341/82,332 rows. CIC mirrors contain 3,119,345 rows; excluding 288,602 unlabeled Thursday-morning extras leaves 2,830,743 labeled rows. Nonfinite and negative numeric values, including unknown-window sentinels, become missing. Remove identifiers and attack category; deduplicate modeled features plus binary label. Earlier roles own repeated feature hashes. Category mappings use training rows only. The manifest records source hashes and exclusions.

UNSW retains 80,000 development and 52,648 holdout rows. Seeds 2-21 split whole development feature groups 60/20/20 into predictor training, calibration and gain fitting. CIC uses label-independent hash samples: 60,000 Monday-Wednesday training, 40,000 Thursday calibration/gain (37.5/62.5) and 40,000 Friday holdout flows. Connected feature/session groups keep related flows and augmented copies together. Models train separately. Friday changes attack families and prevalence relative to Thursday; target labels never recalibrate the detector.

3.2 Frozen comparisons and inference

The histogram-gradient-boosting classifier uses 100 iterations, 15 leaves, learning rate 0.07 and L2 penalty 1. Logistic calibration uses clean, outage and severe copies. Gain heads use 100 iterations, learning rate 0.06 and L2 penalty 5; grouped 80/20 inner validation selects three or seven leaves before refitting. Early stopping is disabled. Budgets are 5%, 10% and 20%.

Field families partition timing, context, volume and related measurements using the manifest lists. Outage removes one uniformly selected family from 50% of records; severe loss removes three families from every record. Plausible replacement substitutes one family on 50% of records using a training-row donor. Generators receive no labels. Calibration/gain augmentation excludes replacement. This is synthetic telemetry degradation; measurement age and adversarial evasion are untested.

Comparators include probability-only, quality in automation only, quality in both heads, observed features, single-head gain, confidence, Chow risk, random and rules-only routing. The conformal adaptation uses 0.05 class-conditional tail budgets, strict order-statistic thresholds fitted on gain rows, and a review band that joins mandatory

cases; excess band cases hold. It is a capacity adaptation of the released procedure [17], not a reproduction of the complete published system. Corrupted calibration copies are dependent and temporal exchangeability is unsupported, so its original finite-sample guarantees are not asserted here.

The primary contrast compares automation-only quality with probability-only estimation on UNSW outage data at 20% capacity, averaging reviewers within each seed. The plan predates fitting. An audit corrected omitted interactions in the observed-feature comparator; all fits were rerun and primary results remained identical. Success requires positive reduction, at least 1% relative improvement and two-sided $p < 0.05$. T-based 95% Monte Carlo intervals describe 20 experimental randomizations on fixed corpora, not new-network uncertainty. The other seven dataset/condition contrasts share one Holm adjustment [33]; remaining comparisons are descriptive.

Report cost, automatic attack closure/all attacks, benign escalation/all benign, held attacks/all attacks, utilization and violations. Unmet attacks include automatic misses, expected reviewer misses and every held attack. Holds never count as resolutions. Sensitivity revalues fixed routes at missed-attack costs 2, 10 and 20 without refitting.

4 Results and discussion

The primary quality benefit was conditional. Cost fell from 0.2144 to 0.2076, a 3.20% reduction (absolute 0.00686; 95% Monte Carlo interval 0.00561-0.00811; $p < 0.001$; 20/20 randomizations). Identical review counts exclude additional analyst effort as the explanation. Table 1 shows lower benign escalation (24.4% to 23.5% of benign flows) and unmet attacks (0.72% to 0.57% of attacks). These percentages refer to the retained benchmark, not operational incident rates.

Temporal replication bounded the conclusion. CIC outage cost changed from 0.6926 to 0.6914 (0.16%; Holm $p = 1.000$). Confidence cost was 0.5193 there and 0.2909 on UNSW. Table 2 retains every declared contrast: UNSW clean/replacement reductions survive Holm adjustment; no CIC contrast does. CIC replacement has an adverse estimate. Clean-UNSW benefit suggests that quality proxies also encode distributional structure; these experiments do not identify the causal effect of repairing telemetry.

Rules exposed capacity and objective limitations. The chosen thresholds flag every severe-loss flow and no injected outage/replacement records. At 20% capacity, approximately 80% of severe flows remain held and the automation-only contrast is exactly zero. Native quality-both ordering leaves 96.9% of severe-condition attacks held on UNSW and 76.0% on CIC. A constant hold price can favor inexpensive reviews over unresolved threats; severity- and delay-dependent prices require domain evidence.

Alternative policies changed the trade-off. On CIC outage data, conformal-adapted cost was 0.8345; it held 9.7% of attacks and 48.4% of all flows. Lower unmet attacks accompanied more unresolved benign traffic. Figure 2 displays both cost and unmet attacks. Removing severe-condition rules automated approximately 80% of policy-

Table 1: Outage results at 20% review capacity, averaged over randomizations and reviewers. Close and Unmet are percentages of all attacks. Conformal adaptation adds band restrictions; other methods share matched counts. Other metrics are in the saved outputs. Means are descriptive.

	UNSW-NB15			CICIDS2017		
Method	Cost	Close	Unmet	Cost	Close	Unmet
Probability only	0.2144	0.06	0.72	0.6926	41.24	48.92
Quality in A	0.2076	0.04	0.57	0.6914	41.19	48.82
Quality in A and H	0.2086	0.05	0.55	0.6785	40.13	47.88
Observed features	0.2077	0.03	0.52	0.6591	38.76	46.50
Confidence	0.2909	0.00	0.05	0.5193	27.18	36.45
Chow risk	0.2116	0.03	0.50	0.6537	37.77	46.17
Conformal adaptation	0.2571	2.83	12.48	0.8345	0.08	10.78

Table 2: Quality-in-automation versus probability-only cost at 20% capacity. Positive differences favor quality. * denotes the primary; Holm covers the other seven rows. MC intervals describe randomizations on fixed corpora. Severe-condition zeros are structural.

Data	Condition	Δ	Rel. (%)	95% MC interval	p	Holm
UNSW	clean	+0.0084	+4.59	[+0.0068, +0.0100]	< 0.001	< 0.001
UNSW	outage*	+0.0069	+3.20	[+0.0056, +0.0081]	< 0.001	–
UNSW	severe	+0.0000	+0.00	[+0.0000, +0.0000]	1.000	1.000
UNSW	replacement	+0.0102	+3.80	[+0.0066, +0.0138]	< 0.001	< 0.001
CIC	clean	+0.0105	+1.67	[-0.0057, +0.0267]	0.191	0.764
CIC	outage	+0.0011	+0.16	[-0.0105, +0.0127]	0.844	1.000
CIC	severe	+0.0000	+0.00	[+0.0000, +0.0000]	1.000	1.000
CIC	replacement	-0.0144	-2.18	[-0.0276, -0.0012]	0.034	0.169

barred flows. Compliance, defect coverage and effectiveness are distinct.

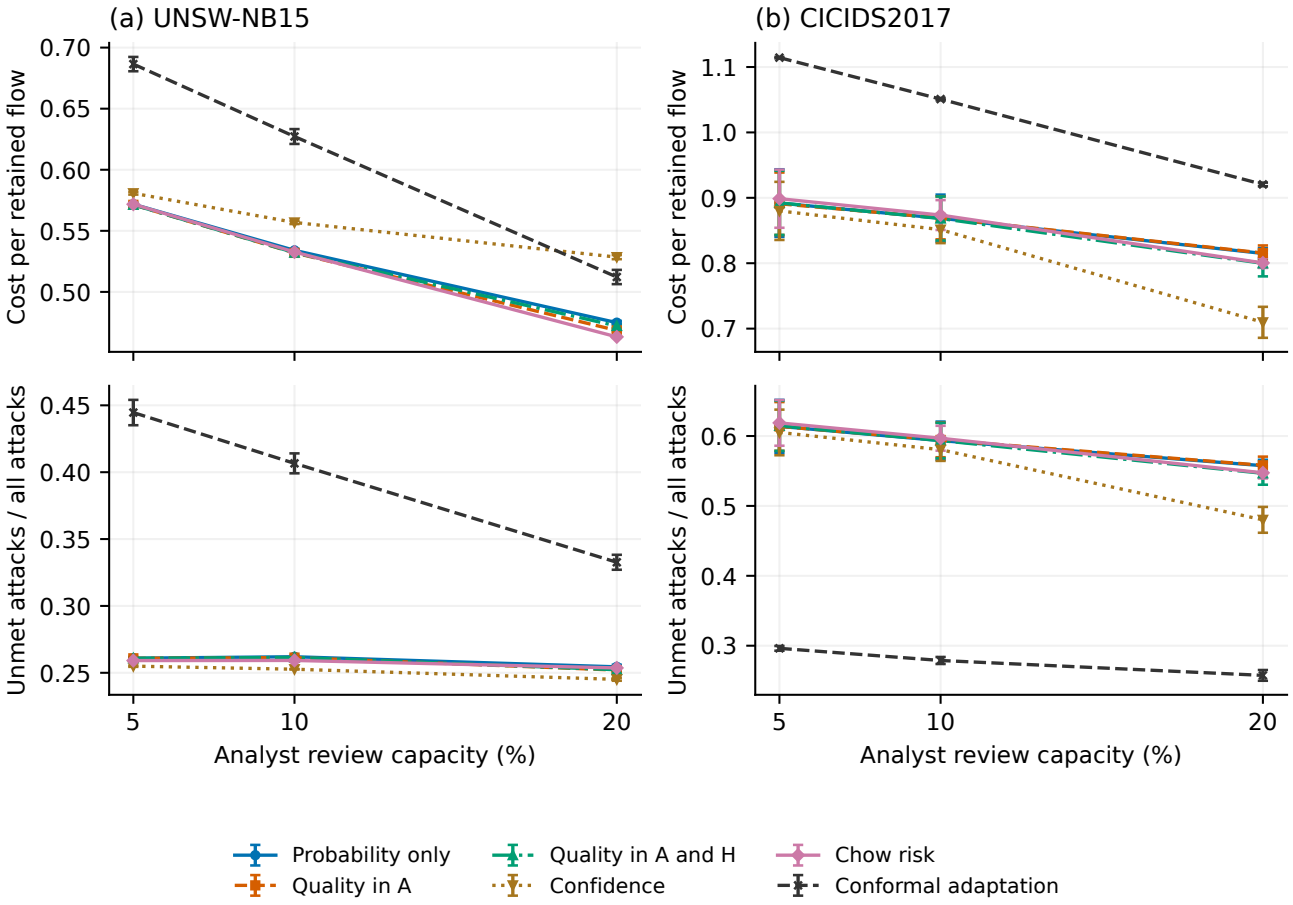


Figure 2: Cost and unmet attacks versus capacity, averaging all four conditions and both reviewers. Bars are 95% Monte Carlo intervals over 20 randomizations on fixed corpora. Shared matched constraints exclude the conformal adaptation, which adds a review band. Generated from saved outputs with AI-assisted coding.

Managers should benchmark quality features against inexpensive heuristics and report unresolved threats alongside automatic errors. UNSW outage savings remain positive when fixed routes are revalued at missed-attack costs 2, 10 or 20. These are illustrative valuations; staffing decisions additionally require arrival rates and investigation times.

5 Limitations and conclusion

The old, controlled corpora and mirror discrepancy limit external validity and provenance assurance. Deduplication, sampling and temporal class shifts alter the evaluated population. Simulated reviewers cannot establish human effectiveness. Real investigation traces, costs and service times remain necessary. Quality proxies certify neither trustworthy telemetry nor security. Fairness, legal compliance, adaptive attackers and containment are untested. L2DHF and DeCCaF are relevant but not newly reproduced numerical baselines. Novelty concerns the controlled

conditions analysis, not a new surrogate or general allocation theorem.

Quality-aware loss estimation met the prespecified UNSW benchmark criterion; temporal CIC evidence did not corroborate its outage benefit. Mandatory overload imposed a structural boundary and exposed the constant hold price. Separating these outcomes supports a testable, bounded contribution to AI governance.

CRediT authorship contribution statement

Yeshwanth Reddy Kunam: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing - original draft, Writing - review and editing, Visualization.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Code, complete aggregate outputs, source verification, analysis plan and environment records are available at `https://github.com/ykunam/data-quality-budgeted-deferral/tree/f5bb01a155bda5ee4f62810ffdc5f39fd61b7e4d`. The initial plan is commit cab4cdec; the comparator amendment is commit e16fe85e. Raw third-party corpora remain at the cited providers and pinned mirrors; they are not redistributed. The release documents access and provenance limitations. A persistent research-data archive remains to be deposited.

Declaration of generative AI and AI-assisted technologies in manuscript preparation

ChatGPT (OpenAI) assisted study design, code, experiments, analysis, drafting and verification. Claude (Anthropic) assisted earlier financial-domain extensions and drafting. Figures were produced programmatically from the implemented design and saved outputs. The author retains responsibility for scientific claims, review of the final code and results, and approval of the submission.

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